

Power Plant Engineering

19-205-0505

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Syllabus

Module Module III

Steam power plants General layout fuel handling systems types of furnaces stokers burning systems types of firing : stokers, pulverized coal burners and fluidized bed combustion power plant boilers, mountings and accessories dust and ash handling systems – draft and chimney calculations – condensers cooling systems - Environmental aspects of thermal power systems Nuclear power plants - Fundamentals of nuclear fission nuclear power plants reactors classification – components layout of simple plant – nuclear power safety and waste disposal.

Module IV

Non conventional energy sources solar radiation and its measurement Solar energy collectors Applications of solar energy Wind energy conversion – site selectio wind energy collectors – Energy from biomass - ocean energy possibilities and future scope – Ocean Thermal electric conversion (OTEC) – Tidal energy - geothermal energy- Magneto Hydro Dynamic (MHD) power– Fuel cells - thermo electric power - thermionic generation.

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3. Morse, Power Plant Engineering, Van Nostrand Co., (1953).
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Power Generation worldwide

Coal	44%
Oil	10
Natural gas	8.5
Nuclear	17
Hydroelectric	20
Other	0.5
Total	100.0%

Steam Power Plants

Rankine Cycle

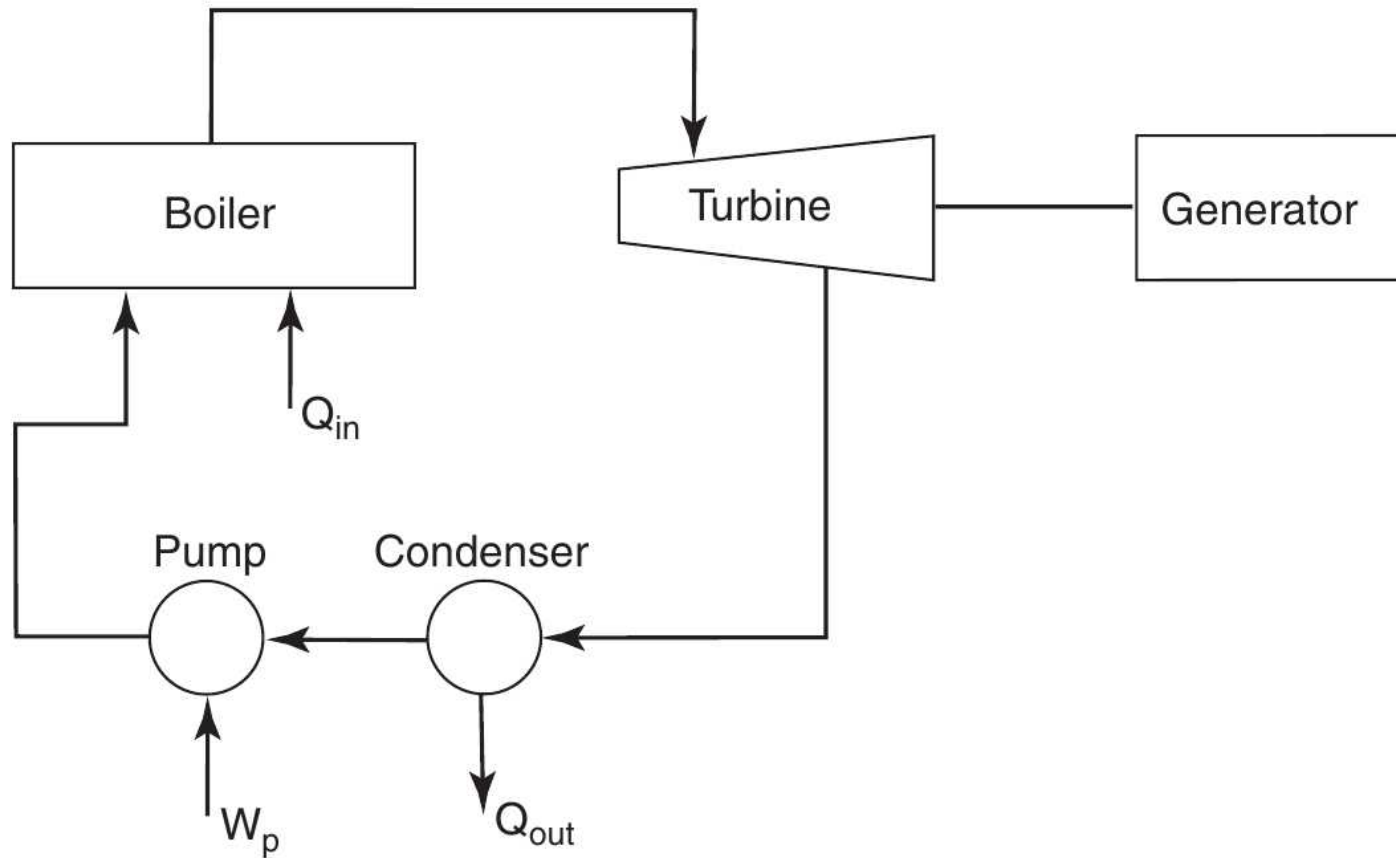


Figure 1.1 Schematic diagram for a Rankine cycle.

Pulverized-coal-fired Power plant systems

1. Coal receipt and preparation
2. Coal combustion and steam generation
3. Environmental protection
4. Turbine generator and electric production
5. Condenser and feedwater system
6. Heat rejection, including the cooling tower

Gas Turbine Power Plants

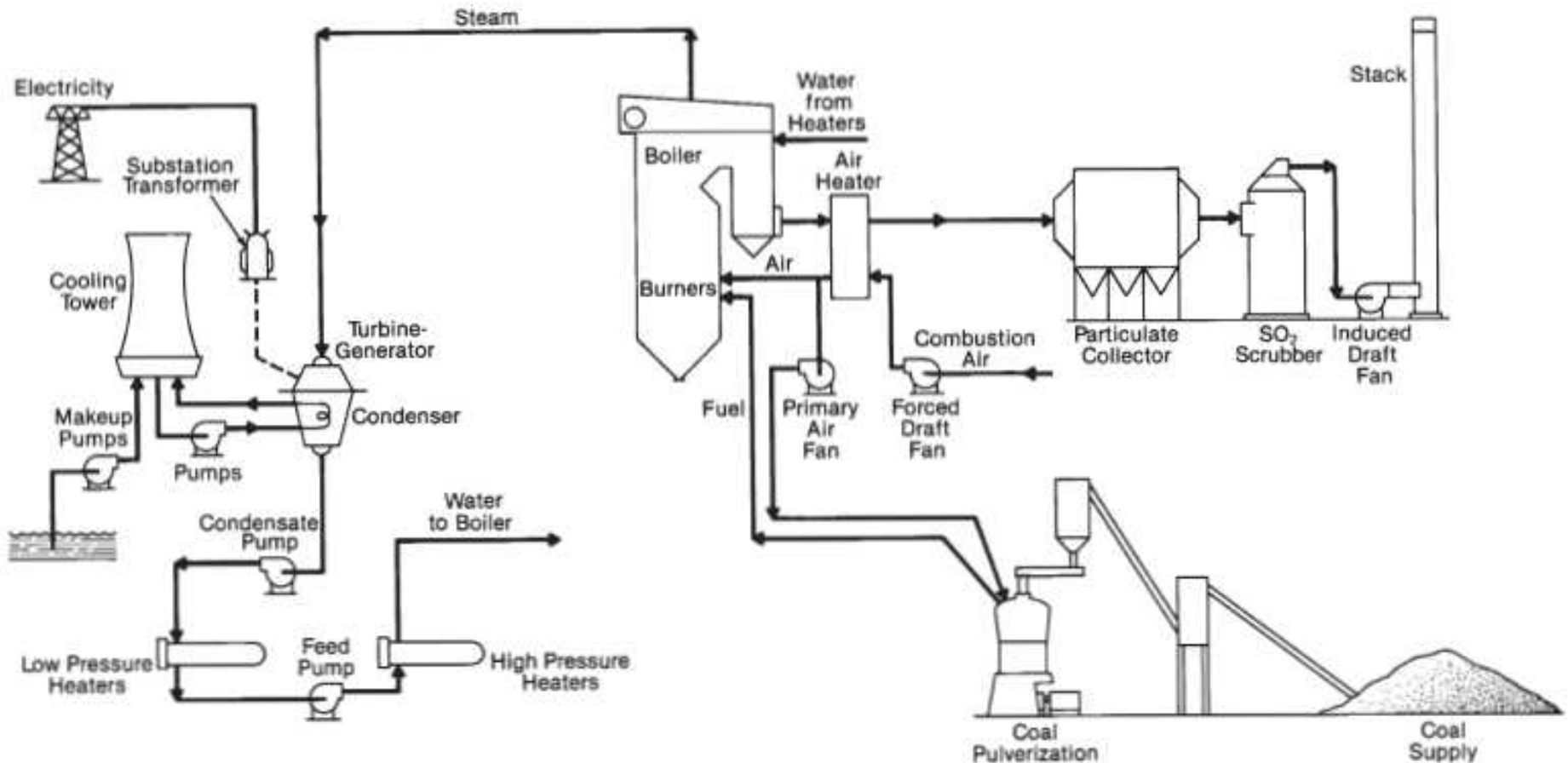


Figure 1.2 Schematic of a typical pulverized coal-fired utility power plant. Reheater, ash and reagent handling, and sludge disposal are not shown. (*Babcock & Wilcox, a McDermott company.*)

Thermal Power Plants

1. Sootblowers, which are special cleaning equipment to reduce the impact of fouling and erosion
2. Air heaters, which provide air preheating to dry fuel and enhance combustion
3. Environmental control equipment such as electrostatic precipitators, bag filterhouses, and SO₂ scrubbers
4. Ash handling systems to collect and remove ash
5. Ash disposal systems including a landfill

Boilers

1. Simplicity in construction, excellent workmanship, materials conducive to low maintenance cost, high efficiency, and high availability
2. Design and construction to accommodate expansion and contraction properties of materials
3. Adequate steam and water space, delivery of clean steam, and good water circulation
4. A furnace setting conducive to efficient combustion and maximum rate of heat transfer
5. Responsiveness to sudden demands and upset conditions
6. Accessibility for cleaning and repair
7. A factor of safety that meets code requirement

Natural Gas and Coal fired Boilers

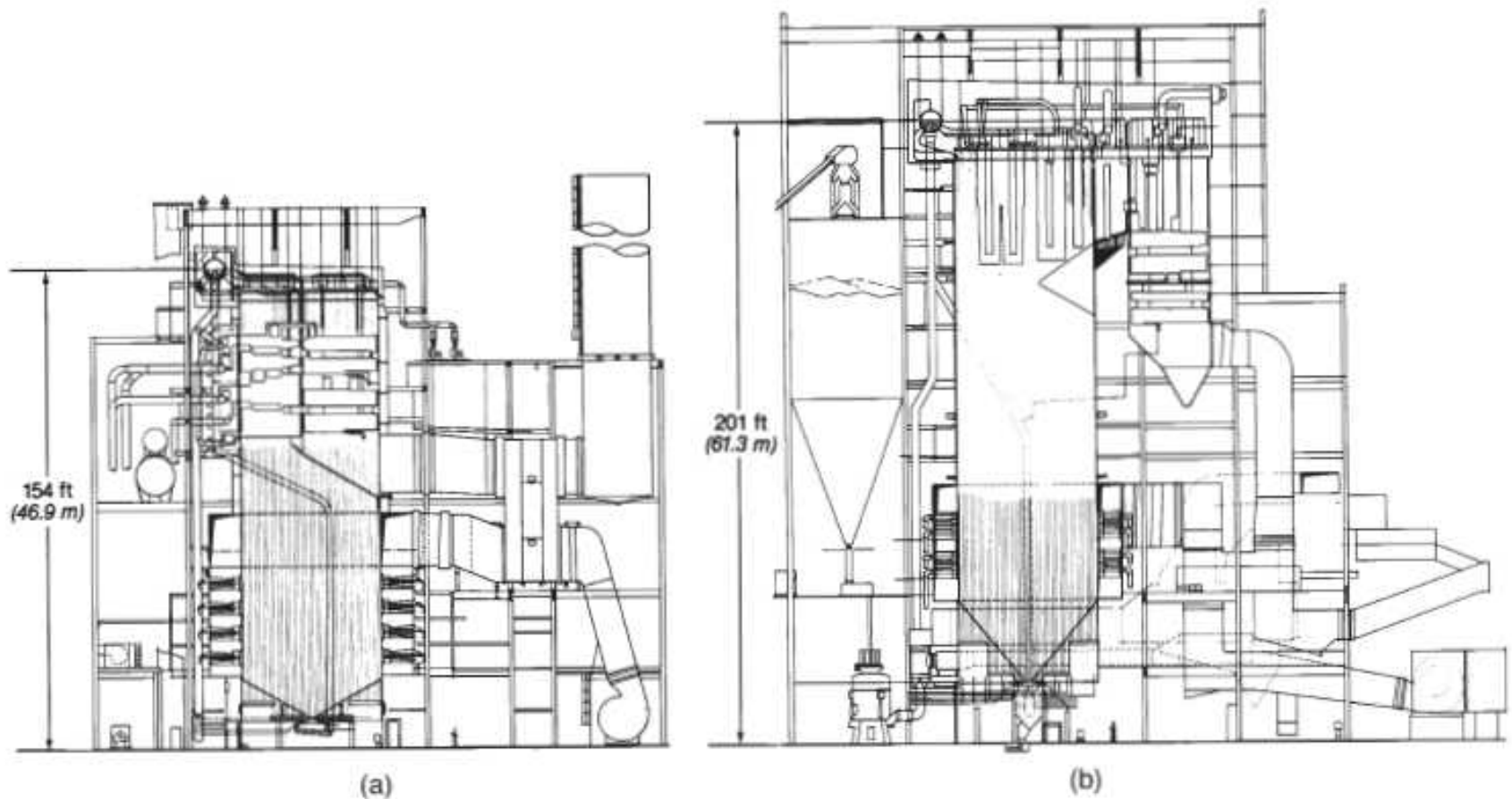


Figure 1.3 Comparison of (a) a natural gas-fired boiler and (b) a pulverized coal-fired boiler, each producing the steam at the same capacity, pressure, and temperature. (*Babcock & Wilcox, a McDermott company.*)

Furnaces

Furnace: It is a structure under or adjacent to a boiler in which fuel is burned and from which the combustion products pass into the boiler. It provides a chamber in which the combustion reaction can be isolated and confined so that the reaction remains a controlled force. It provides support or enclosure for the firing equipment.

Furnaces may be classified ;

- In terms of a construction feature
- Operating characteristics
- General shape
- Design purpose
- The fuel they burn

Stokers

- Stokers are designed to feed solid fuel onto a grate.
- The fuel burns as primary air is passed up through it.
- Over-fire air is also introduced to enhance combustion.
- The stoker is located in the furnace
- It also designed to remove the residue of ash after combustion.

Stoker firing consists of

1. A fuel feed system
2. A grate assembly, either stationary or moving, that supports the burning fuel and admits the majority of combustion air to the fuel
3. An over-fire air system for the completion of combustion and reduction of emissions such as NO_x
4. An ash discharge system

Stokers

Two types of Stokers;

1. Underfeed stokers: Both fuel and air supply are from under the grate
2. Overfeed stokers: The fuel is supplied from above the grate and air is supplied from below the grate.

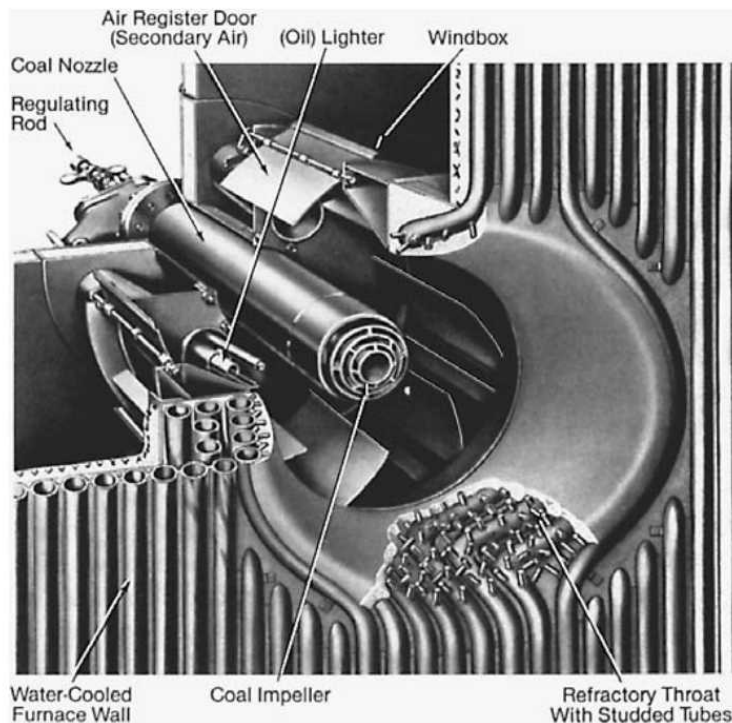


Figure 5.32 Circular register burner for pulverized coal firing. (Babcock & Wilcox, a McDermott company.)

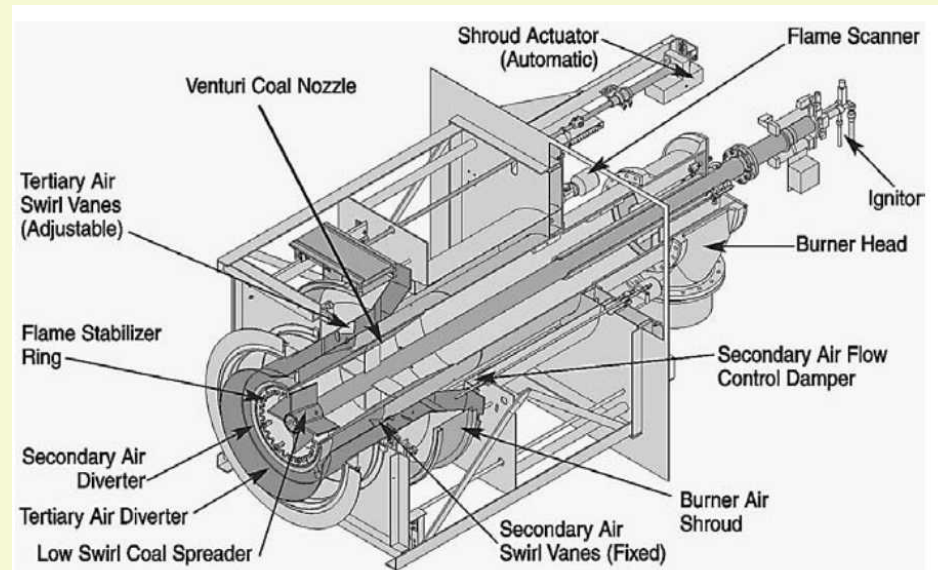
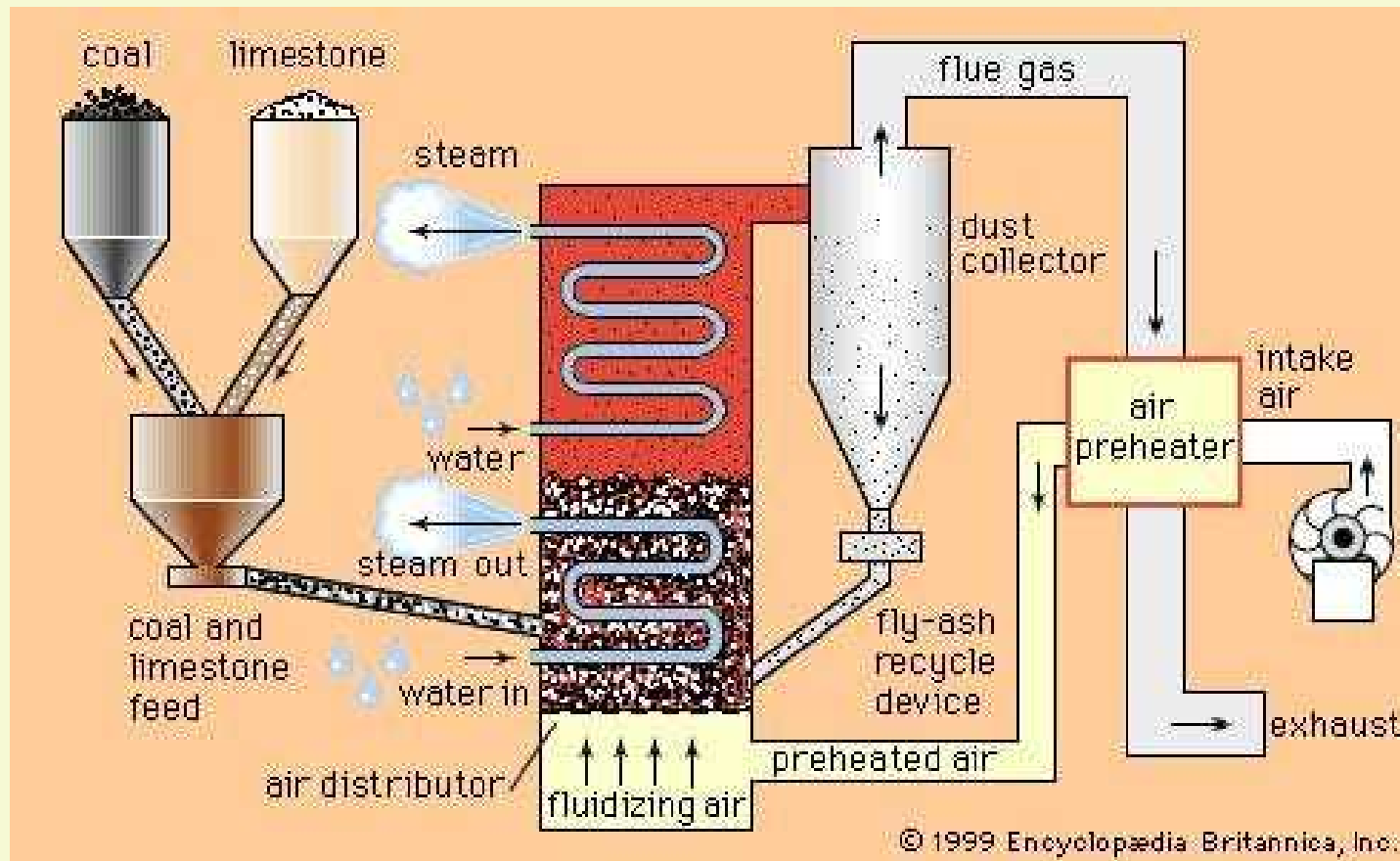


Figure 5.33 Controlled combustion venturi (CCV) low- NO_x dual air zone burner for the burning of pulverized coal. (Riley Power, Inc., a Babcock Power, Inc., company.)

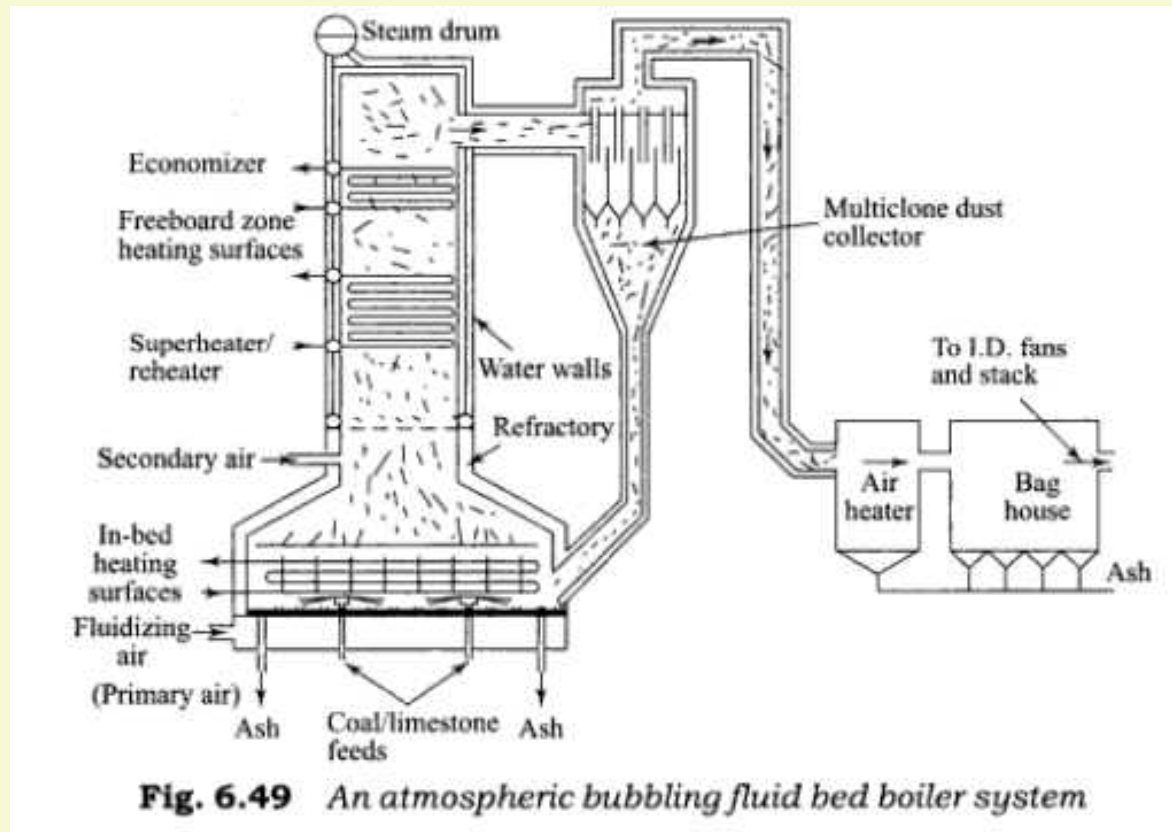
Fluidized bed combustion Boilers

Fluidized bed combustion (FBC) is a specialized combustion process wherein solid particulates are suspended in upward jets of air in order to achieve more effective chemical reaction and heat transfer.



Fluidized bed combustion Boilers

- Coal or biomaterials, are suspended in a stream of air or gas, forming a “fluidized” bed of particles.
- The bed behaves like a fluid, hence the name “fluidized bed.”
- This process improves combustion efficiency and heat transfer.
- Particles are constantly in motion and have a large surface area, allowing for better mixing of fuel and oxygen.



Boiler Accessories

- Water Columns
- Pressure Measurement
- Temperature Measurement
- Feedwater Regulators
- Safety Valves
- Blowdown Equipment-blow down the boiler periodically or even, at times, continuously. Blowdown therefore is the water that is bled from the boiler drum to control the concentration of total solids in the boiler water.
- Nonreturn Valves
- Steam Piping
- Sootblowers
- Valves

Ash Handling Systems

- Ash handling systems are designed for the removal of ash from boilers burning coal or other solid fuels.
- These systems handle bottom ash and fly ash from stoker-fired, pulverized coal-fired, and fluidized bed boilers.

Type of ash	Utility boilers	Industrial boilers
Bottom ash		
Dry	Hydraulic or mechanical	Hydraulic, mechanical, or pneumatic
Molten	Hydraulic or mechanical	Hydraulic or mechanical
Fly ash	Pneumatic	Mechanical or pneumatic
Economizer and air heater ash	Hydraulic, mechanical, or pneumatic	Mechanical or pneumatic

Ash Handling Systems

Submerged Chain conveyor Ash handling system

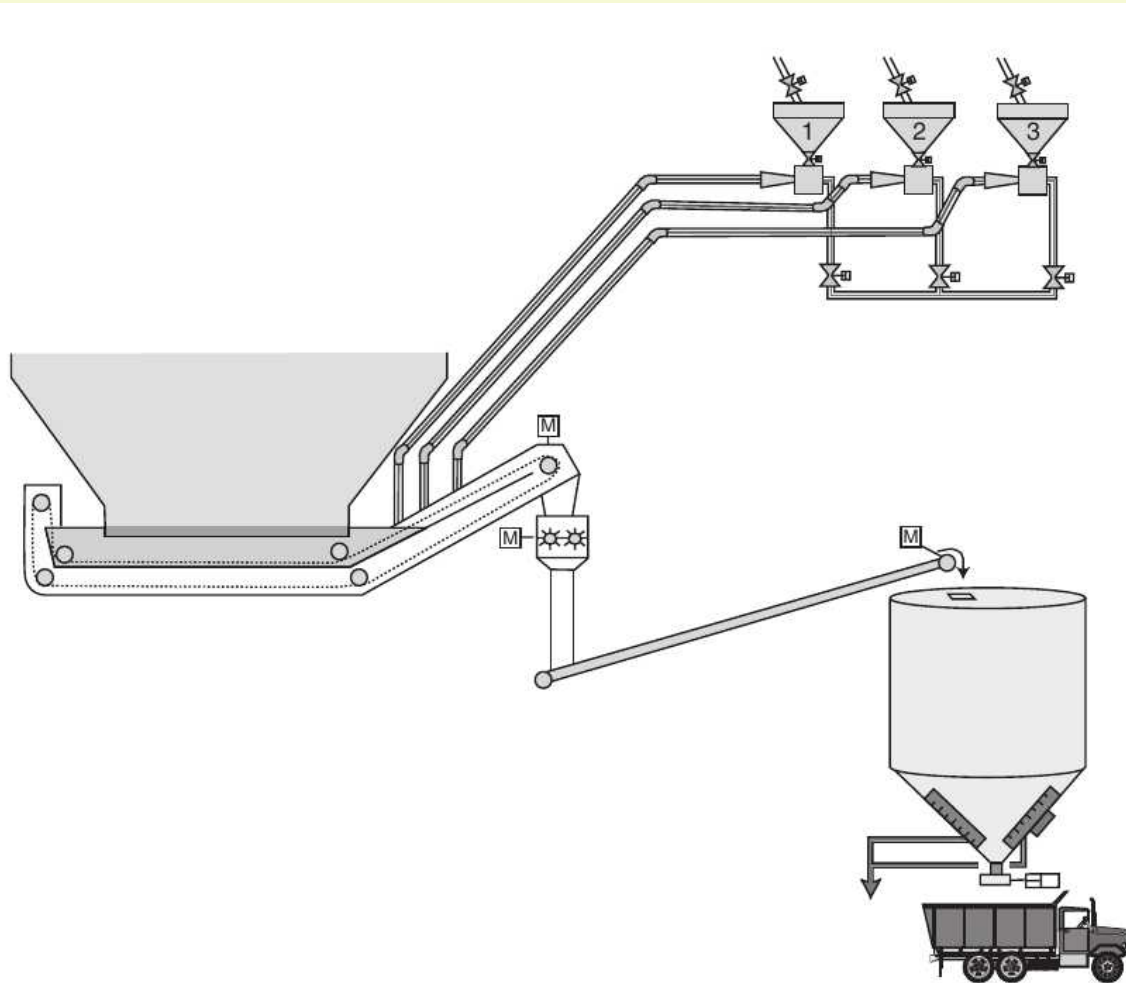
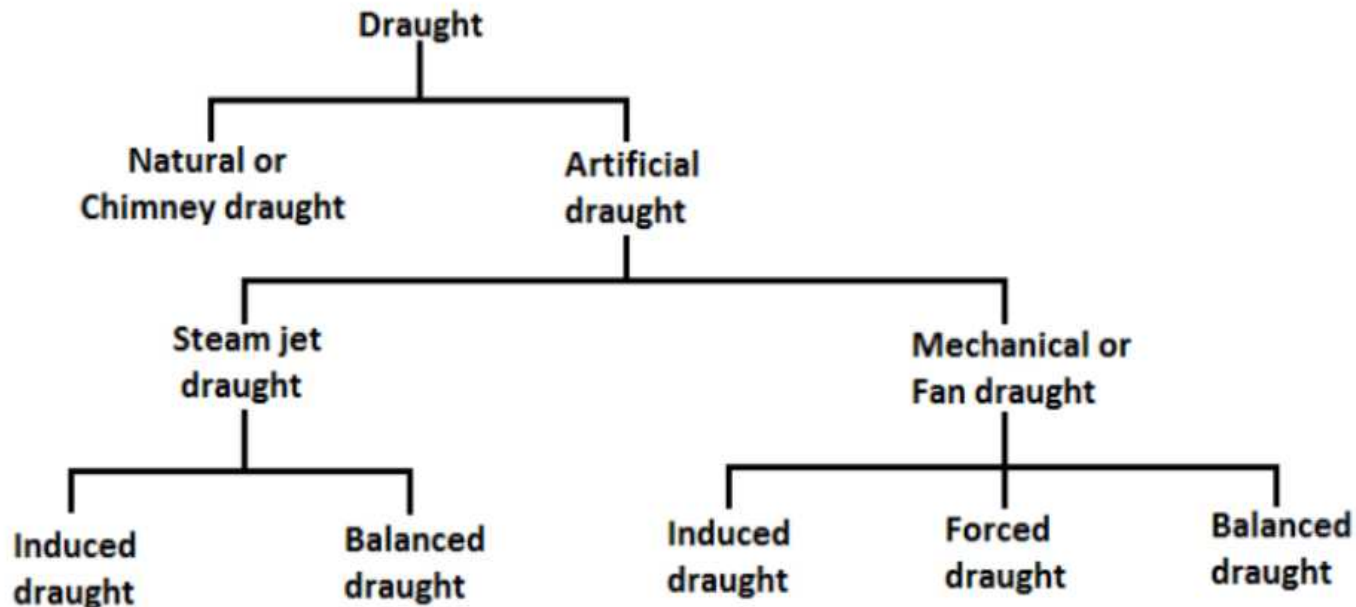


Figure 5.52 Submerged-chain conveyor type of ash-handling system. (Allen-Sherman-Hoff, a Division of Diamond Power, Inc.)

Boiler draught

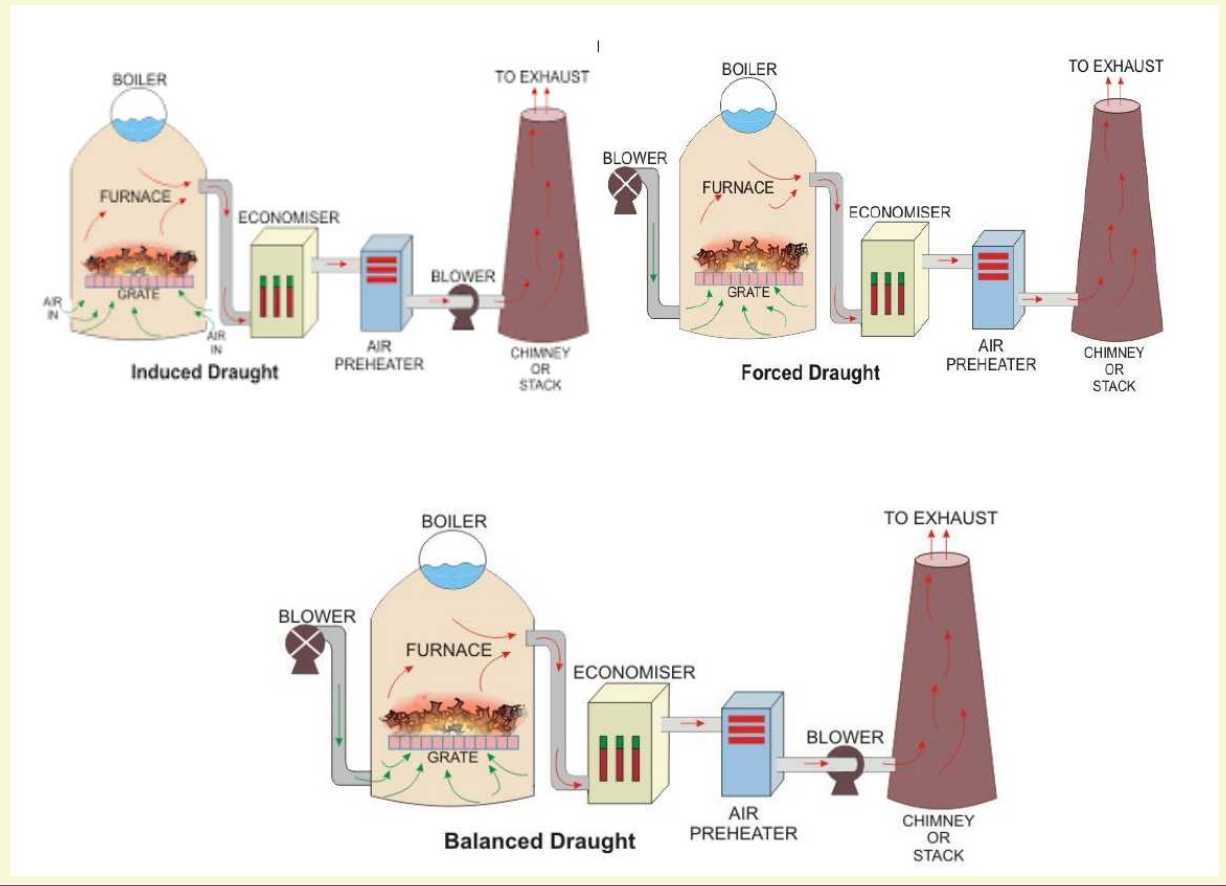
- Boiler draught may be defined as the small difference between the pressure of outside air and that of gases within a furnace or chimney at the grate level, which causes flow of air/hot flue gases to take place through boiler.



Classification of Draught

Purpose of Boiler Draught

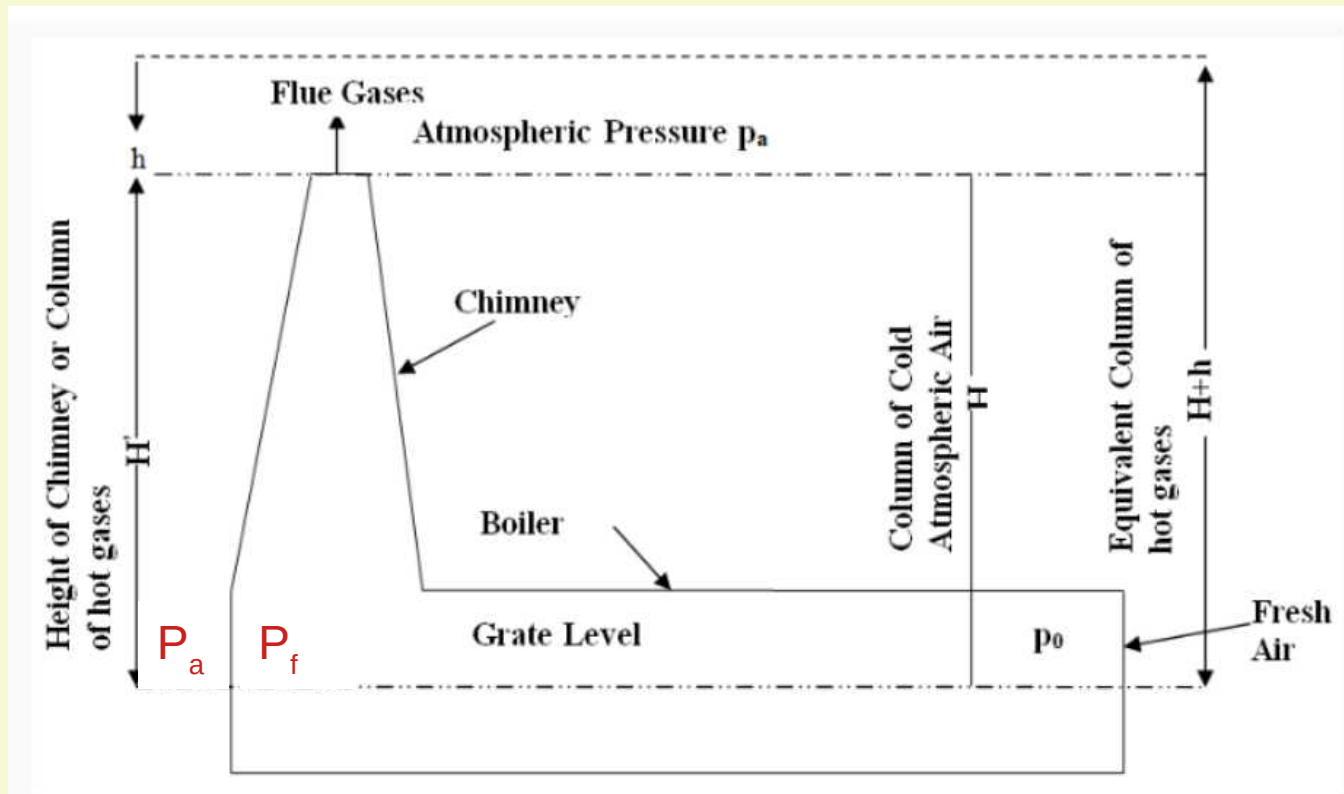
1. To provide an adequate supply of air for fuel combustion.
2. For throw out the exhaust gases of combustion from the combustion chamber.
3. To discharge these gases to the atmosphere through the chimney.



Natural Draught

Advantages of Natural Draught

- It does not require any external power for producing the draught.
- Initial investment is less.
- The maintenance cost is low as there is no mechanical part.



Natural Draught calculation

$$P_a - P_f = \Delta P \text{ (N/m}^2\text{)} = (\rho_a - \rho_f)gH$$

at NTP, $T_0 = 273 \text{ K}$, $P_0 = 101.3 \text{ kN/m}^2$, $R = 0.287$, $v_0 = \frac{RT_0}{P_0}$

$$v_0 = \frac{0.287 \times 273}{101.325}, v_0 = 0.7732 \text{ kg/m}^3$$

$$V_0 = mv_0 \quad \text{and} \quad \frac{V_a}{T_a} = \frac{V_0}{T_0}$$

volume of air with temperature T_a

$$V_a = 0.7732m \times \frac{T_a}{273}$$

$$\rho_a = \frac{m}{V_a} = \frac{273}{0.7732 \times T_a}$$

For the flue gas

$$m_f = 1 + m$$

$$\rho_f = \frac{m_f}{V_f} = \frac{273}{0.7732 \times T_f}$$

$$\Delta P = 353gH \left(\frac{1}{T_a} - \frac{1}{T_f} \right)$$

In terms of water column (mm) $h = 353H \left(\frac{1}{T_a} - \frac{1}{T_f} \right)$

Natural Draught calculation

Chimney diameter

Velocity of flue Gas $V = \sqrt{2 g H}$

Flow rate of flue Gas = $\rho_f A_c V$

Area of the Chimney $A_c = \Pi D^2/4$

- Importance of Chimney Height
- Centralised Chimneys
- Individual Chimneys



Mechanical Draught

- Forced Draught fans
- Induced Draught fans
- Balanced Draught

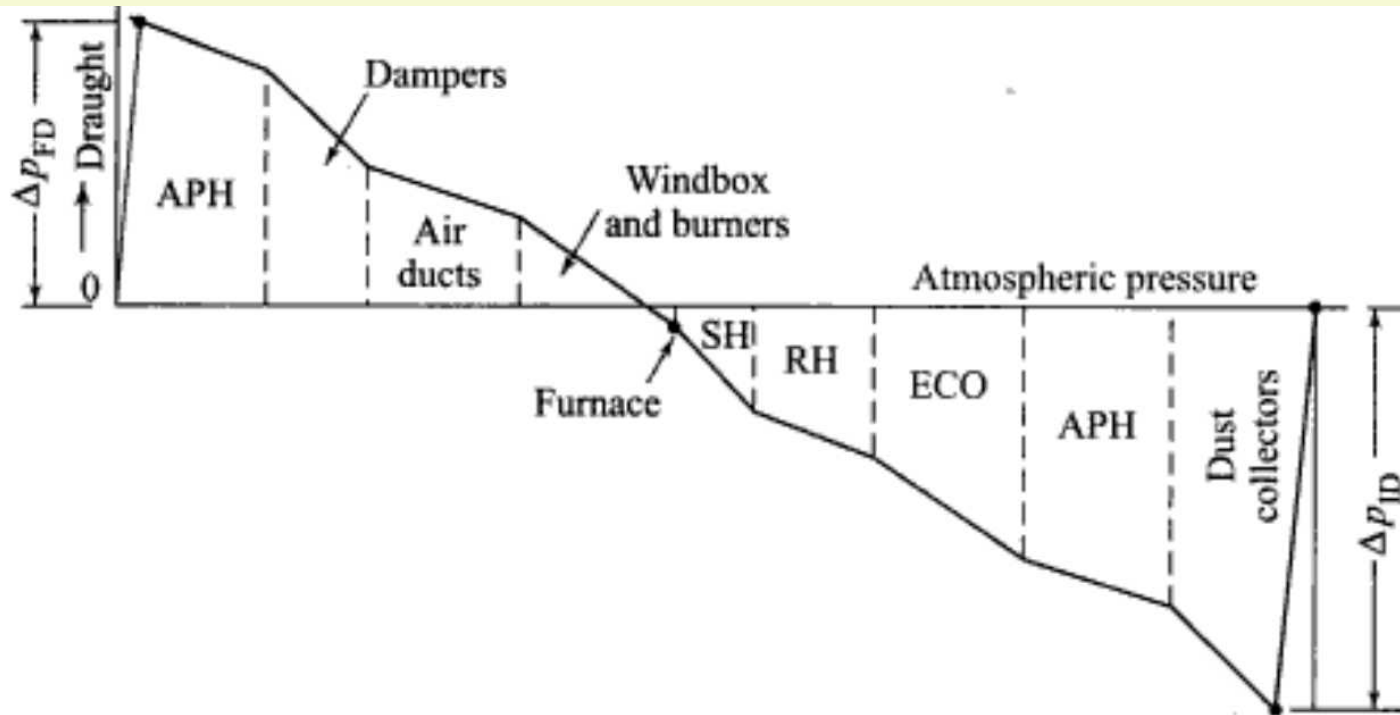
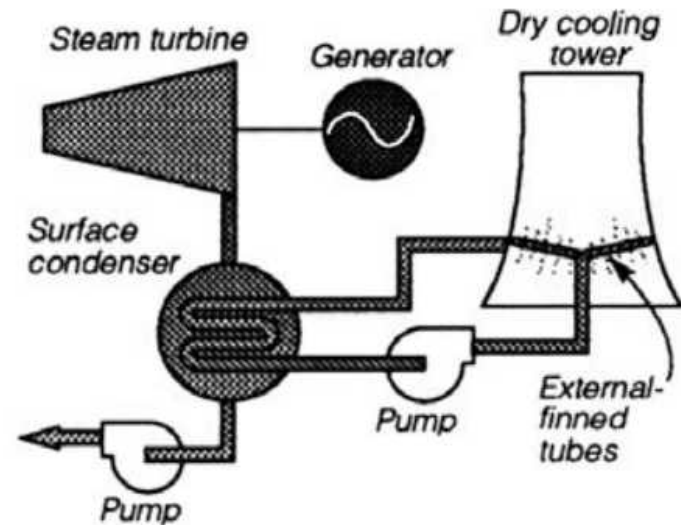
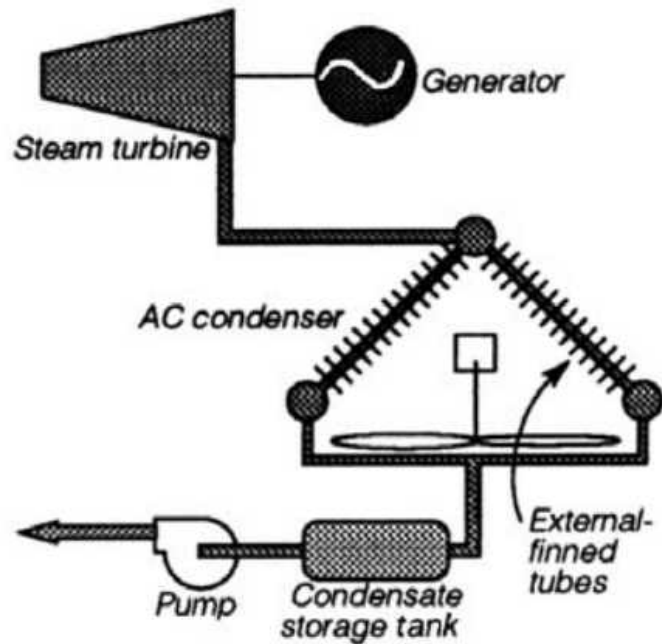


Fig. 4.16 *Balanced draught system in a modern boiler*

Condensers

- To condense the steam exhaust from the turbine
- Heat rejected to atmospheric Air/Water using heat exchangers.
- Shell and Tube Heat Exchangers, Air Cooled condensers, Cooling Tower etc,



Environmental aspects of Thermal Power Plants

Major pollution contributing factors

- **Carbon dioxide**
 - **0.82 kg of CO₂ /kWhr generation (Approx, India)**
 - **India total electricity consumption 2023 - 1407 (Twh), (India 70% Thermal power)**
- **Carbon monoxide**
- **NO_x**
- **SO_x**

Environmental aspects of Thermal Power Plants

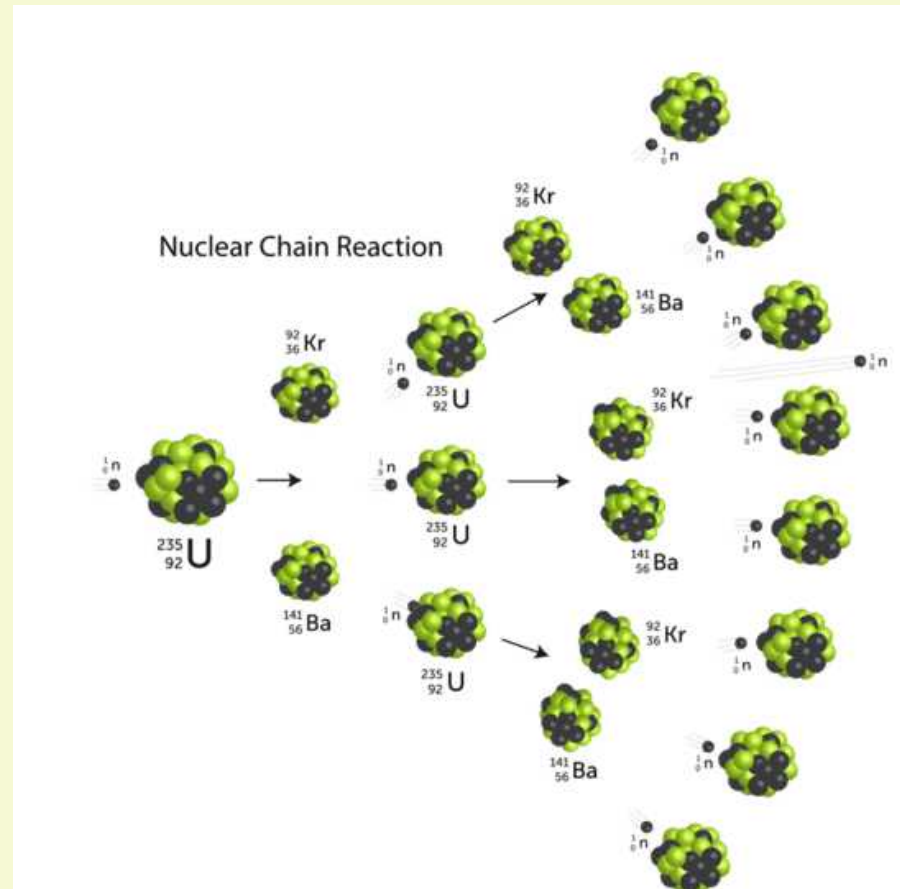
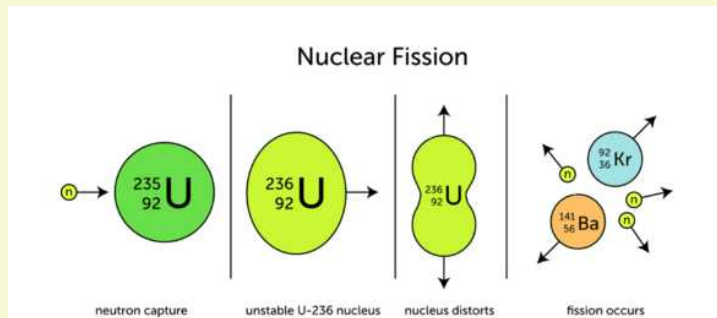
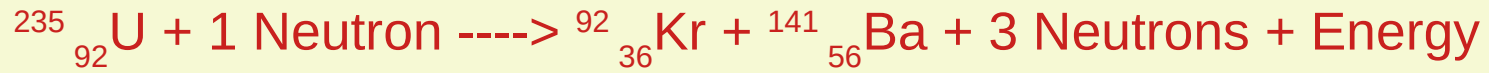
- **Particulate matter (PM)**

Particulate matter is the sum of all solid and liquid particles suspended in air, many of which are hazardous

- Water pollution
- Land pollution
- Noise pollution

Nuclear Power Plants

Nuclear Fission



Nuclear Power Plants

Nuclear Fission

Moderators- Slow down fast moving neutrons

- High purity graphite, Beryllium, Reactor-grade heavy water

Control rods - Control the rate of fission in the nuclear fuel

- They are made from materials that absorb neutrons, such as boron, cadmium, hafnium, silver, or indium

Nuclear Power Plants Classification

Neutron Flux Spectrum	Moderator	Coolant	Fuel material
Thermal	Light water	Light Water	Enriched uranium
	Heavy water	Heavy water	Natural uranium
	Graphite	Gas (CO ₂)	Natural or enriched uranium
Fast	Nil	Liquid metal (Na, K)	Plutonium, Thorium

Nuclear Power Plants

Pressurised Water Reactor

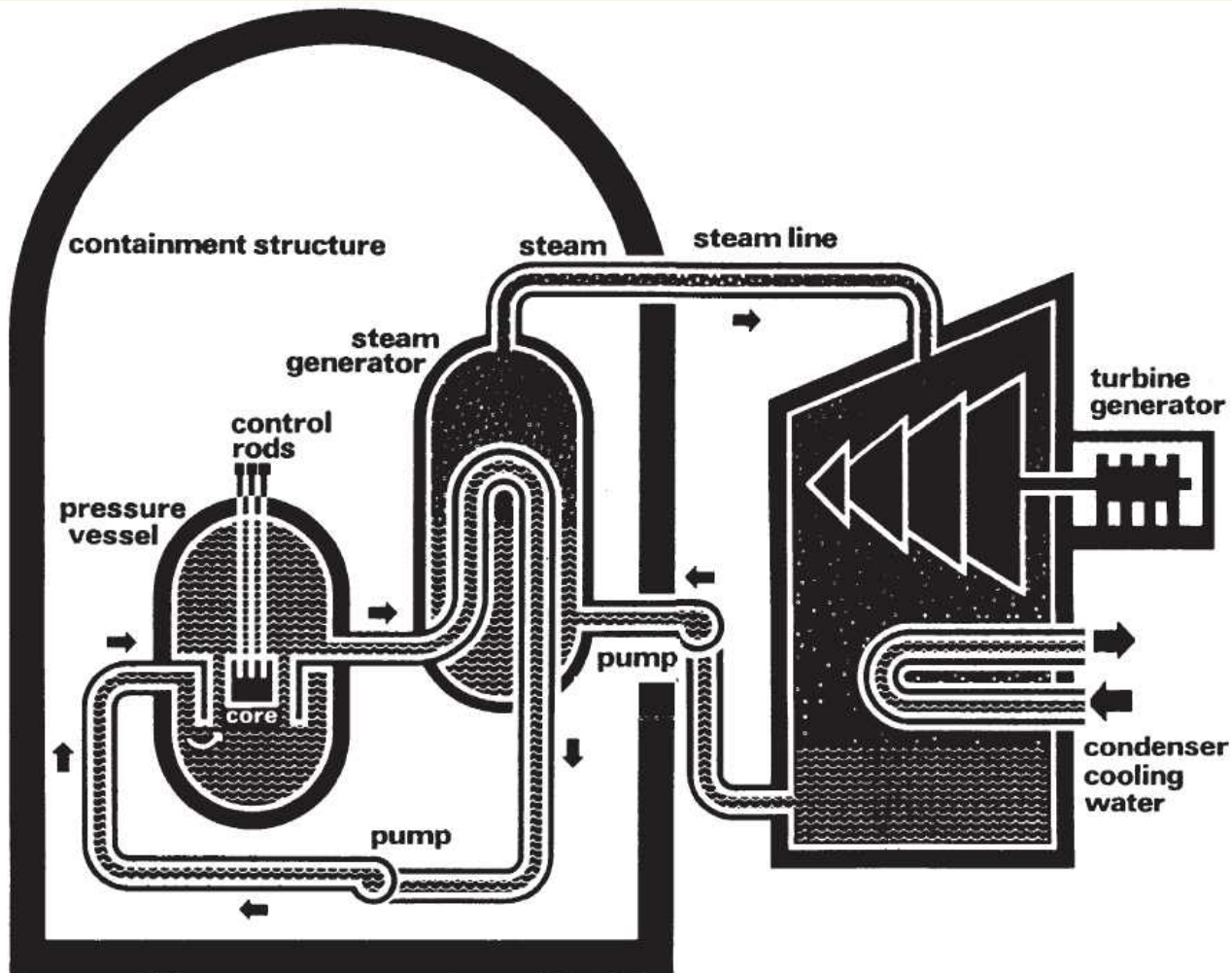


Figure 2.57 Pressurized-water reactor (PWR) nuclear power plant. (*American Nuclear Society.*)

Nuclear Power Plants

Fast Breeder Reactor

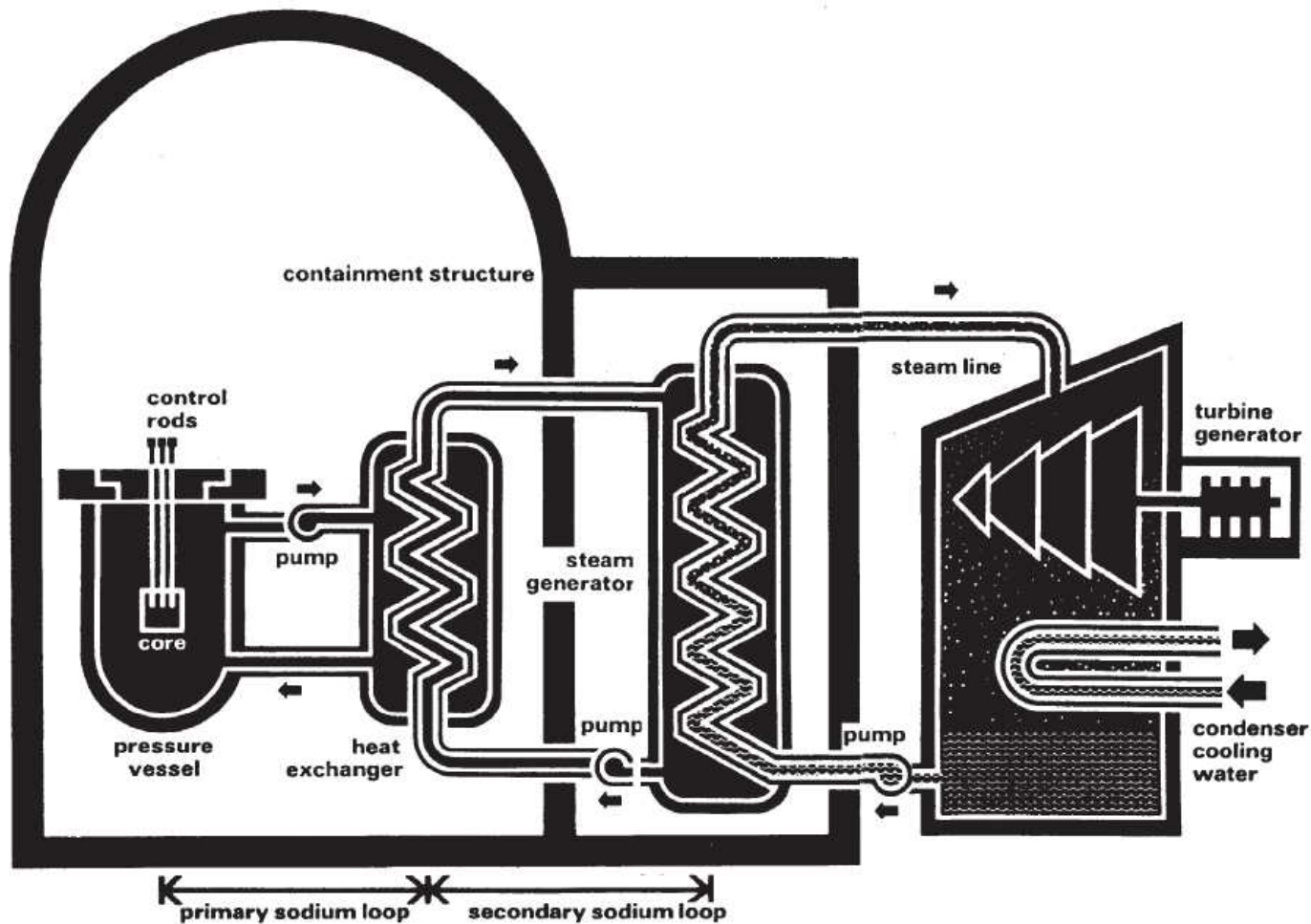


Figure 2.59 Liquid-metal fast breeder reactor (LMFBR) nuclear power plant. (*American Nuclear Society.*)

Nuclear Power and the Environment

Nuclear Fuel cycle

1. Mining
2. Milling and refining U_3O_8
3. Processing to produce Uranium Hexafluoride
4. Isotopic enrichment of Uranium Hexafluoride to reach reactor requirement
5. Fabrication of reactor fuel elements
6. Power generation in the reactor
7. Waste management

Nuclear Power and the Environment

Nuclear Fuel cycle Waste

1. Gaseous effluent-released to environment – Biosphere and dispersed into atmosphere slowly
2. Uranium mine and mill tailings- Residues with low levels of natural radiation Stored in the mines itself
3. Low level wastes- Contain less than 10 nano Curies per gram- Disposed in liquid form in shallow land burial
4. High level wastes- Generated during reprocessing fuel-
5. Spent fuel- Removed in fuel structural stored under water for 3-4 months before disposal.